MIL-N-00494B
26 October 1984
USED IN LIEU OF
MIL-N-494A
22 March 1963

MILITARY SPECIFICATION

NITROGUANIDINE

This limited coordination military specification has been prepared by the US Army Armament, Munitions and Chemical Command based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of MIL-N-494A. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers nitroguanidine (NQ) for use in propellants.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-I-1795	-Ink, Marking, Stencil, Opague (Porous and Non-Porus Surfaces)
PPP-T-60	- Tape, Packaging, Waterproof

FSC 1376

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-B-117	-	Bag, Sleeve and Tubing - Interior Packaging
MIL-B-46506	-	Box Ammunition Packing, Wood, Wirebound
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-F-50449	-	Filler, Sheet Form (For Use In Ammunition Containers)

STANDARDS

FEDERAL

FED-STD-101	-	Preservation and Packing Materials: Test Procedures

MILITARY

MIL-STD-109	-	Inspection Terms and Definitions
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-286	-	Propellants Solid: Sampling, Examination and Testing
MIL-STD-1171	-	Acceptance and Description Sheets (For Propellants and Explosives)
MIL-STD-1218	-	ACS Chemicals
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1233	-	Procedure for Determining Particle Size Distribution and Packing Density of Powdered Material

FEDERAL

FED-STD-101	-	Preservation and Packing Materials: Test Procedures
TT-I-1795	-	Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces)
PPP-T-60	-	Tape, Packaging, Waterproof

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the document which is indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto if applicable.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E300 - Recommended Practice for Sampling Industrial Chemicals
ASTM D2905 - Statement on Number of Specimens Required to Determine the Average Quality of a Textile Material

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA. 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

CODE OF FEDERAL REGULATIONS

49 CFR 100-199 - Department of Transportation Code of Federal Regulations Title 49

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 100-199 (latest revision)".)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Raw materials used to manufacture nitroguanidine shall be in accordance with applicable specifications and the approved process control document.

3.2 Purity. The purity of the nitroguanidine will be 99.00 percent minimum, when determined by the method specified in 4.5.1.

3.3 Ash content. The ash content shall be 0.30 percent maximum, when determined by the method specified in 4.5.2.

3.4 pH value. The pH value shall be greater than or equal to 4.5 but less than or equal to 7.0 when determined by the method specified in 4.5.3.

3.5 Acidity. The acidity (measured as percent sulfuric acid) shall be 0.06 percent maximum when determined by the method specified in 4.5.4.

3.6 Total volatiles. The total volatiles shall be 0.25 percent maximum when determined by the method specified in 4.5.5.

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3.7 Sulfates. The sulfates (measured as percent sodium sulfate) shall be 0.20 percent maximum when determined by the method specified in 4.5.6.

3.8 Water insoluble impurities. The water insoluble impurities shall be 0.20 percent maximum when determined by the method specified in 4.5.7.

3.9 Average particle size. The average particle size of the nitroguanidine shall be greater than or equal to 3.4 microns but less than or equal to 6.0 microns when determined by the method specified in 4.5.8.

3.10 Color. Nitroguanidine shall be a white color when determined in accordance with 4.5.9.

3.11 Consistency. Nitroguanidine shall be a free flowing, crystalline powder when determined in accordance with 4.5.10.

3.12 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 6.2).

3.13 Process controls. The contractor shall submit a process control document to the government specifying the process variables which are considered critical for the production of nitroguanidine. The process control document shall be submitted in accordance with 4.1.1.

3.14 Workmanship. The manufacturer shall implement procedures and controls to assure that the process and the nitroguanidine product are not compromised by foreign materials and contaminant or any other conditions which may degrade the nitroguanidine. Compliance with this requirement shall be as specified in 4.5.11.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. In addition, the government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein.

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4.1.1 Submission of process control data. A process control document shall be submitted to the Technical Agency (DRSMC-QAR-R(D)) through the contracting officer, in accordance with Data Item Description, DI-P-1614 (Tailored). The document shall contain a description of the process, all raw materials used, critical process conditions/procedures and production/inspection equipment used to produce nitroguanidine meeting the requirements of this specification (see 6.2.2). In addition, whenever there is a change (change in source of materials, critical process conditions, procedures, etc.) from previously reported process control data, the Technical Agency shall be notified of the new process control data and the reason for the change. The Technical Agency should be notified at least 4 weeks in advance of executing any planned changes in the process control document.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

c. Packaging inspection (see 4.4.2 and 5).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall be ten pounds of nitroguanidine. The sample shall be obtained from the first production lot which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be performed. The first article sample may be subjected by the government to any or all of the examinations and tests specified in this specification (see Table I).

4.3.3 Rejection. If the first article sample fails to comply with any of the applicable requirements, it shall be rejected. The government reserves the right to terminate its inspection upon any failure of the first article sample to comply with any of the stated requirements.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots will be formed in accordance with MIL-A-48078 and inspected in accordance with MIL-STD-1235.

4.4.2 Examination. The unit of product for the examinations specified in 4.4.2.1 through 4.4.2.4 below shall be the individual drum or box. Alternative sampling plans may be used by the contractor if they provide, as a minimum, the protection afforded the government by the applicable sampling plans in MIL-STD-1235 and if they are approved by the procuring activity.

TABLE I. First article inspection

	REQUIREMENT PARAGRAPH	MIL-N-00494B
Purity	3.2	4.5.1
Ash Content	3.3	4.5.2
pH Value	3.4	4.5.3
Acidity (as sulfuric acid)	3.5	4.5.4
Total Volatiles	3.6	4.5.5
Sulfates (as sodium sulfate)	3.7	4.5.6
Water Insoluble Impurities	3.8	4.5.7
Average Particle Size	3.9	4.5.8

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Box, wooden, prior to sealing (see 5.1.1)		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined.				
Major 101	Liner in box missing or inadequate		.40%	Visual	
Minor 201	Liner in box cut, torn or punctured		1.00%	Visual	
202	Insufficient packing material		1.00%	Visual	

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Box, wooden, sealed (see 5.1.1, 5.1.2 and 5.2)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined.				
Major					
101	Box damaged		.25%	Visual	
Minor					
201	Contents loose		1.50%	Manual	
202	Marking misleading or unidentifiable		1.50%	Visual	
203	Evidence of poor workmanship (see 3.14)		1.50%	Visual	4.5.11
NOTES:					

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Fiber drum, prior to sealing (see 5.1.1, 5.1.2, and 5.1.3)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
Critical	None defined.				
Major					
101	Foreign matter in nitroguanidine (see 3.14)		.40%	Visual	4.5.11
102	Check color of nitroguanidine (see 3.10)		.40%	Visual	4.5.9
103	Check consistency of nitroguanidine (see 3.11)		.40%	Visual	4.5.10
Minor	None defined.				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Fiber drum sealed (see 5.1.1, 5.1.2, and 5.2)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
Critical	None defined.				
Major					
101	Drum damaged		.25%	Visual	
102	Improper assembly (cover not flush, tape not as specified)		.25%	Visual	
Minor					
201	Marking misleading or unidentifiable		1.50%	Visual	
202	Evidence of poor workmanship (see 3.14)		1.50%	Visual	4.5.11

NOTES:

DD FORM 1664 (8) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.4.3 Testing. The unit of product for the tests listed in Table II and described in section 4.5 of this specification shall be one batch of nitroguanidine, if a batch production process is used, and one shift's production of nitroguanidine, if a continuous production process is used.

4.4.3.1 Testing frequency. The units of product shall be selected for inspection in accordance with MIL-STD-1235, CSP-1 Plan, AQL 6.5%, code letter B. The units of product will be sampled in accordance with paragraph 4.4.3.2 below. If any unit of product fails to meet any test requirement, the unit of product represented by the sample shall be rejected. All units of product produced between the time that the last unit of product was tested and accepted and the unit of product which failed shall be tested for the requirement which failed in accordance with the applicable test method given in paragraph 4.5. If any of these units of product fail to meet the subject requirement, that unit of product shall also be rejected. In addition, after any failure of a unit of product for a specific requirement, the contractor will return to 100% inspection for that requirement until "i" successive units of product are accepted as required by MIL-STD-1235. During this period, reduced testing (according to MIL-STD-1235) is still required on all other characteristics at the frequency specified in MIL-STD-1235.

4.4.3.2 Sampling from units of product. Two drums shall be selected at random from each unit of product selected for sampling. A single sample of about one pound of nitroguanidine shall be taken from each drum in accordance with the sampling procedure for solids in ASTM E300. Each sample of nitroguanidine will be analyzed in duplicate by the methods given in section 4.5 of this specification to ascertain conformance with the requirement of section 3 of this specification.

4.4.4 Classification of defects. The classification of defects shall be as given in Table II.

TABLE II. Classification of defects

Purity (see 3.2)	Major defect
Ash content (see 3.3)	Major defect
pH value (see 3.4)	Major defect
Acidity (see 3.5)	Major defect
Total volatiles (see 3.6)	Major defect
Sulfates (see 3.7)	Major defect
Water insoluble impurities (see 3.8)	Major defect
Average particle size (see 3.9)	Major defect

4.4.5 Inspection equipment. For the performance of all tests and examinations specified in 4.4 and 4.5, commercial inspection equipment should be employed. The contractor shall

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have available, and utilize correctly, this equipment and is charged with the responsibility of assuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes (see 6.4).

4.5 Methods of inspection. All tests given in this section shall be performed using prescribed procedures for replicate determinations given in standard analytical chemistry textbooks or ASTM D2905. Also, unless otherwise specified herein, all chemicals shall be Reagent Grade or ACS Grade in accordance with MIL-STD-1218. See 6.5 for use of equivalent test methods. The tests (First Article and Quality Conformance) listed in the item detail specification shall be performed on inspection lots as defined in 4.4.1 of this specification.

PRECAUTION: This specification covers sampling and testing of toxic and hazardous material. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing of the nitroguanidine.

4.5.1 Determination of purity. The purity shall be determined by either the nitrometer method or the alternative ferrous sulphate potentiometric titration method. The method used shall be reported with the result.

4.5.1.1 Nitrometer method. The nitrometer method specified in MIL-STD-286, method 209.3 shall be used as the standard method for purity except that a 3 gram sample shall be crushed so that it will pass through a US Standard Number 100 sieve. A 1 gram sample shall be dried in air at 46° Centigrade (°C), overnight and then placed in an oven for 1 hour at 100°C. The sample shall then be transferred to the nitrometer. The percent nitrogen shall then be converted to percent nitroguanidine by the following formula:

$$\text{Percent Nitroguanidine} = \frac{7.430R}{W}$$

Where: R = reading of measuring tube
W = sample weight
7.430 = conversion factor from %N to %NQ

4.5.1.2 Ferrous sulphate potentiometric titration method.

4.5.1.2.1 Purging the system. If the system has been down for at least 8 hours, purge the system. Set the selector switch on the burette/dispenser to manual dispensing of titrant and let approximately 100ml or more of titrant flow into a 250ml beaker. Dispose of the spent titrant into the reserve titrant solution storage bottle for reuse in the next solution(s). Set the selector switch on the burette/dispenser to remote.

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4.5.1.2.2 Zeroing the system. Prior to operating the system, the electrodes should have a minimum of 4 hours conditioning in a sulfuric acid bath. Unplug the electrodes and p'ug in the shorting plug. Turn the function switch from standby to 1000 mv. Adjust the standardize control until the instrument is exactly at zero. Pull the zero-pull switch and adjust the voltage to 1000mv by adjusting the zero fine/course control. (WARNING: Do not turn on switch with electrodes connected. This will put a charge on the electrodes and cause an error in the readings. Unplug the shorting plug and plug in the electrodes.)

4.5.1.2.3 Standardization of the system using potassium nitrate (KNO_3).

Weigh a sample bottle containing 0.5000 gm $\pm$ 0.0500g of KNO_3 that has been dried in a 135°C oven for a minimum of 4 hours. If the KNO_3 has been out of the oven for four hours or more, redry in a 100°C oven for a minimum of 2 hours. Turn on the pump to the constant temperature unit to start water circulating through the bath. The constant temperature unit should be set at 35 $\pm$ 2°C.

Place the magnetic stirring bar into a dry hot 250ml beaker (or titrator cup) and add 150ml of sulfuric acid (H_2SO_4). Place the beaker in the bath on the stirrer unit. With the stirrer operating, slowly pour the sample into the center of the swirl. Be careful not to touch the pipet or electrodes with the weighing bottles. Lower the electrodes and the pipet into the H_2SO_4. Set controls at the desired settings (see Table III). Turn the function switch to 1000mv. Set the timer for five minutes. Reweigh the weighing bottle to find the weight of the standard by difference. Push the start switch on the burette/dispenser unit and the unit will begin to dispense ferrous sulphate ($FeSO_4$) into the beaker. Manually adjust the speed of titration by manipulating the proportioning band control to produce an overall titration time of 5 to 8 minutes. Only experience will dictate how to adjust the titration rate since it will continuously have to be changed at the beginning of the titration. The titration at the end point should be as slow as possible to get a good end point (see Table III).

After the end point is reached and has held steady for at least one minute, turn the Function switch to standby and push the Fill switch. It is important to push the Fill switch. If it is not pushed, the unit may start titrating when the electrodes are lifted from the beaker. There is a possibility that acid could be splashed on the units and the operator. Also if the Fill switch is not pushed the system will start titrating when the electrodes are lowered into the acid on the next sample.

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Raise the electrodes from the acid and carefully wipe by dabbing the electrodes and pipet with a tissue. Discard the tissue into a container with water. Remove the beaker containing the spent titrating solution (and titrated KNO_3) from the cooling bath.

Rinse the electrodes and pipet with clean sulfuric acid and let the excess acid on the electrode drain into the rinse beaker while preparing acid and sample for the next analysis. Record the ml reading from the digital readout meter. Calculate the nitrogen equivalence factor (F) for the standard KNO_3 by the equations given below:

$$F = \frac{A \times 13.853}{B}$$

Where: A = weight of the KNO_3 standard
13.853 = % N in KNO_3
B = ml of $FeSO_4$ titrant required to titrate the standard KNO_3

4.5.1.2.4 Analysis of Nitroguanidine (NQ.) A 3.00 gram sample shall be crushed so that it will pass through a US Standard Number 100 sieve. Prepare a sample of NQ weighing 0.5000g $\pm$ 0.0500g. Repeat the procedure as given in section 4.5.1.2.3 for standardization of the system using KNO_3 with the following exceptions noted below:

4.5.1.2.4.1 Use the instrument settings given in Table III for the NQ sample.

4.5.1.2.4.2 The NQ sample should be dried in an oven at 45°C for 16 hours and then in a 100°C oven for 1 hour.

4.5.1.2.4.3 The Proportioning Band is set on 100 for NQ and is not changed during the titration as in the KNO_3 procedure.

4.5.1.2.4.4. The timer shall be set at five (5) minutes for NQ. This dissolving time must be exact.

The purity of the NQ sample is found from the equations given below:

$$\% \text{ Purity} = \frac{C \times F \times 7.4299}{D}$$

Where: C = ml of $FeSO_4$ titrant required to titrate the sample of NQ
F = equivalence factor calculated from the standardization of the KNO_3
D = weight of NQ sample

7.4299 = Conversion factor from %N to %NQ

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TABLE III. Instrument settings

Burette/Dispenser Mode	Variable	Electrometer Temperature	Function
REM	OFF	35°C	(Standby) STBY

Titrate Demand

Endpoint 1	Endpoint 2
200 (NQ)	0000
350 (KNO_3)	

	MV	Titrate				Burette	Proportioning Band-Min Delivery	
EP-1	EP-2	pH	EP-1	EP-2	Ext	Stop Start	+ -	
In	Out	In	In	Out	Out	Out	Out *Variable, See 4.5.1.2.3	100(NQ) 0(KNO_3)

*Final Proportioning Band Setting (guide line only)

80-85 for KNO_3
100 for NQ

4.5.1.2.5 Preparation of ferrous sulphate solution. Dissolve 350 grams of $FeSO_4.7H_2O$ in 800 milliliters of water. Add 1000 ml of a solution made up of equal parts of concentrated sulfuric acid (95-98%) and water. Add water until the final volume is 2.0 liters.

4.5.2 Determination of ash content. The ash content shall be determined as specified in Standard MIL-STD-286, method 106.1.2.

4.5.3 Determination of pH value. A portion of 5.0 grams (gms.) of the sample, weighed to the nearest milligram (mg) shall be transferred to an 8 ounce bottle which has been previously heated to 80°C. A 200 milliliter portion of freshly boiled distilled water, which has been allowed to cool to 80°C shall be added. The bottle and contents shall be immersed in a constant temperature water bath, maintained at 80° ± 2°C. The bottle shall be removed for shaking at 3 minute intervals. As soon as the solution is complete, the bottle shall be removed and rapidly cooled to room temperature in cold running tap water or an ice bath. The pH of the solution shall be determined using a glass electrode pH meter. This solution shall be retained for the "Determination of Acidity" (see 4.5.4). Prior to the pH determination, the meter shall be calibrated using standard buffer solution at pH 8.0.

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4.5.4 Determination of acidity. The solution retained from the determination of the pH value shall be titrated with a 0.05 N sodium hydroxide solution to a reading of 7.6 on the pH meter. A blank determination shall be conducted on an equal volume of distilled water and the volume of sodium hydroxide used for the specimen shall be corrected. The percent acid present in the solution shall be calculated as sulfuric acid as follows:

$$\text{Percent sulfuric acid} = \frac{4.9\ (A-B)\ N}{W}$$

Where: A = ml of sodium hydroxide solution used for titrating specimen
B = ml of sodium hydroxide solution used for titrating blank
N = normality of sodium hydroxide solution
W = weight of specimen in gms

4.5.5 Determination of total volatiles. A portion of the sample, approximately 5.0 gms, weighed to the nearest 0.1 mg, shall be transferred to a tared aluminum dish (90 millimeters (mm) in diameter and 60mm deep) having a tightly fitting cover. The uncovered dish containing the specimen shall be heated for 2 hours in an oven maintained at 100° ± 2°C. At the end of this period the dish shall be covered, transferred to a desiccator until cool, and then weighed. The procedure of heating, cooling in a desiccator, and weighing shall be repeated at hourly intervals until 2 successive weighings do not differ by more than 0.2 mg, or the specimen increases in weight. The percent total volatiles shall be calculated as follows:

$$\text{Percent total volatiles} = \frac{(A-B)\ 100}{A}$$

Where: A = weight of specimen in gms prior to heating
B = weight of specimen in gms after heating

4.5.6 Determination of sulfates. A portion of the sample, approximately 5.00 gms, weighed to the nearest 0.1 mg, shall be dissolved in 200 milliliters of hot distilled water contained in a 400-milliliter beaker. If necessary the solution shall be filtered through a No. 42 Whatman filter paper, or equal. The solution shall be made acid by the addition of 1 milliliter of concentrated hydrochloric acid, which shall be followed by the addition of 1 milliliter of 10 percent barium chloride solution. The solution shall be boiled for approximately 5 minutes and then allowed to stand, with occasional gentle stirrings, until cool. The solution shall then be allowed to stand for at least 10 hours. At the end of this period, the solution shall be heated to boiling and filtered through a previously ignited and tared Gooch crucible equipped with fiberglass filter paper. The precipitate shall be washed with small portions of warm distilled water,

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sucking each portion through before adding the next. The last few washings shall be tested with 1 percent silver nitrate solution to see that all chlorides have been removed. When free of chlorides the Gooch crucible shall be transferred to an oven maintined at 100° ± 2°C, for a period of two hours. The crucible shall then be transferred to a desiccator until cool, and weighed. The percent sulfate present in the specimen shall be calculated as sodium sulfate as follows:

$$\text{Percent sodium sulfate} = \frac{60.86A}{W}$$

Where: A = weight of residue in gms
W = weight of specimen in gms

4.5.7 Determination of water insoluble impurities. A portion of the sample of approximately 10.0 gms, weighed to the nearest 0.1 gm, shall be dissolved in 400 milliliters of boiling distilled water. The solution shall be filtered through a tared Gooch crucible equipped with fiberglass filter paper. The residue shall be washed with a total of 100 milliliters of boiling water, in 20 milliliter portions, letting each portion go through the crucible before adding the next. The crucible shall be transferred to an oven, maintained at 105° ± 5°C for 2 hours, cooled in a desiccator and weighed. The gain in weight shall be calculated as percent water insoluble impurities as follows:

$$\text{Percent water insoluble impurities} = \frac{(A-B)\ 100}{W}$$

Where: A = weight of crucible and residue in gms
B = weight of crucible in gms
W = weight of specimen in gms

4.5.8 Average particle size. The average particle size shall be determined in accordance with MIL-STD-1233, method 100.

4.5.9 Color. Visually observe the production of nitroguanidine and examine the samples submitted for testing with the requirement given in 3.10.

4.5.10 Consistency. Visually observe the production of nitroguanidine and examine the samples submitted for testing with the requirement given in 3.11.

4.5.11 Workmanship. Visually observe the production of nitroguanidine and examine the samples submitted for testing for compliance with the requirements given in 3.14.

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5. PACKAGING

5.1 Packing.

5.1.1 Level A (packing not for tactical deployment). Fifty pounds of nitroguanidine shall be packed in a fiber drum complying with the requirements of 5.1.1.1. After the drum has been filled, the cap (cover) and the body (shell) seam shall be secured with 2 inch wide, minimum, tape complying with PPP-T-60. The drum shall be placed in a barrier bag constructed from material conforming to Type 1, Class F, MIL-B-117. The bag shall be heat sealed. The bagged drum shall be placed in a tight fitting, wirebound box conforming to Type I, Class 1, Grade A, MIL-B-46506. The box shall be lined with mandatory fillers conforming to MIL-E-50499. Use additional fillers as required for tight pack.

5.1.1.1 Fiber drum. The All Fiber Drum, capacity approximately eighteen gallons, shall comply with specification DOT 21C60 and the following requirements:

a. The inside dimensions of the drum shall be approximately 14 inches in diameter and 27 inches in height.

b. The drum shall be of double wall construction. The side wall (outer shell or telescoping cover and inner tube) shall consist of two cylindrical units, convolutely wound and with no single ply less than 0.012 inch thick. Each unit shall have all plies firmly bonded together with a container grade thermoplastic (hot melt) adhesive that will not permit a ply edge separation greater than 1/4 inch after testing for water resistance (ply separation) by method 3028 of FED-STD-101 as suggested for weather resistant fiberboard. All areas more than 1/4 inch from the edges shall remain bonded.

c. The combined plies making up the sidewall (outer shell plus inner tube) shall have a minimum thickness of 0.108 inch and a minimum dry bursting strength (Mullen) of 900 PSI.

d. The drum top and bottom (headings) shall consist of multiples of fiberboard with outer plies containing a polyethylene moisture barrier having a maximum water vapor transmission rate of 1.0 gram/100 sq inches/24 hours at 1 atmosphere, 100°F and 90% RH. The drum headings shall be laminated with water resistant adhesive that will not permit a ply edge separation greater than 1/4 inch when tested by the method used for sidewall described above in 5.1.1.1b.

e. The drum shall have a telescoping cover for flush cap to body (shell). The gap (seam) between the outer sidewall elements (cover and shell) of the assembled drum shall be 0.025 inch minimum to 0.250 inch maximum.

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f. The cover (drum cap) and bottom shall be wax dipped, one inch minimum.

g. The drum shall be fabricated without staples. No metal shall be used in the construction of the drum.

h. The outer construction seams of outer sidewall elements in the cover and shell shall be covered with a 2 inch water resistant tape glued with a water resistant adhesive.

5.1.2 Level B. Level B packing shall be the same as Level A, except the barrier bag and mandatory fillers shall be eliminated however, continue to use fillers as required for tight pack.

5.1.3 Level C. Level C packing shall be the same as Level A except the barrier bag and wirebound box shall be eliminated.

5.2 Marking

5.2.1 Level A

5.2.1.1 Unit packaging (Fiber Drum) shall be marked in accordance with MIL-STD-129. Stencil ink applied by the silk screen method and meeting the performance requirements for Type II, TT-I-1795 may be used for marking.

5.2.1.2 Exterior pack (Wirebound Box) shall be marked on the side and top with letters 1/2 inch (nominal) high, unless otherwise specified, using stencil ink, Type I or III, TT-1-1795 with the following information:

"HIGH EXPLOSIVES - DANGEROUS" (DOT proper shipping name)
"50 POUNDS"
"NITROGUANIDINE"
"NSN 1376-00-628-3329" (1 Inch nominal)
"LOT NO. " (insert lot no. 1 inch nominal)
"GROSS WT. " (insert gross weight in lbs to nearest lb)

5.2.1.2.1 Special Marking. Exterior pack shall be marked on the top and side, with letters 1 inch (nominal) high, using stencil ink, Type I or III, TT-I-1795, as follows: "PACKING NOT FOR TACTICAL DEPLOYMENT".

5.2.2 Level B. Level B shall be marked the same as cited in paragraph 5.2.1.2.

5.2.3 Level C. Level C shall be marked as cited in paragraph 5.2.1.2 except the DOT proper shipping name will be placed within 3 inches of the top and bottom of the drum, and "DRUM NO", in 1 inch nominal letters, may be added below "GROSS WT".

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6. NOTES

6.1 Intended use. Nitroguanidine is to be used in the manufacture of propellants.

6.2 Ordering data.

6.2.1 Procurement requirements. The procurement document should specify the following:

a. Title, number and date of this specification.

b. Quantity required and delivery schedules.

c. First article requirements - sample provisions (see 3.12 and 4.3).

d. Level of packing protection required.

e. The contractor shall submit a first article report which shall contain all necessary identification information relating to the First Article, such as contract number, nomenclature, department of defense ammunition code (DODAC) number 1376-T002, contractor, facility, and test results in quantitative terms for the test and the derived values. The contractor will indicate the specification, drawing or other test equipment that was used as a basis for the examination or test.

6.2.2 Data requirements. When this specification is used in a procurement contract which incorporates DD Form 1423, acceptance and description sheets prepared in accordance with MIL-STD-1171, DRSAR-P-702-109 or as directed in the contract shall be supplied by the contractor. In addition, the Process Control Document (see 3.1, 3.13, and 4.1.1) shall be submitted in accordance with DI-P-1614 (tailored).

6.3 International standardization agreement. Certain provisions of this specification are the subject of International Standardization Agreement, STANAG No. 4026, and ABC-ARMY-STD-114. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that the appropriate action may be taken respecting the international agreement concerned.

6.4 Submission of inspection equipment designs for approval. Submit equipment designs, as required to: Commander, ARDC, AMCCOM, ATTN: DRSMC-QAR-R(D), Dover, New Jersey 07801.

6.5 Equivalent test methods. Prior approval of the contracting officer is required for use of equivalent test methods. A description of the proposed method should be submitted

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thru the contracting officer to: Commander, ARDC, AMCCOM, ATTN: DRSMC-QAR-R(D), Dover, NJ 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
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(Project 1376-A289)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
IL-N-00494B	NITRGUANIDINE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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